

Framed Formal Monodromy of Cubic Threefolds

Tavis Rudd

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Abstract

We define a framed formal-monodromy count ν_6 for the numerical small even quantum connection and compute $\nu_6(X) = 2$ and $\nu_6(X \times \mathbf{P}^1) = 4$ for every smooth cubic threefold. The calculation and product formula are unconditional. Under an explicit reconstruction-tail hypothesis, the count satisfies blowup and projective-bundle formulas; under a separate residual divisor-tagging hypothesis for the remaining surface centers, it is birationally invariant through dimension four. These hypotheses give a conditional second proof of one-step irrationality and the conditional identity $\nu_6(V) = 2$ for prime Fano threefolds of genus eight. We isolate the exact comparison statements still needed to make the refinement unconditional. A strictly admissible specialization on F_1 shows why the tagging hypothesis must be restricted to actual center maps.

Keywords. cubic threefold; quantum D -modules; formal monodromy; primitive sixth roots; blowups; projective bundles.

1 Introduction

The direct formal-exponent marker of [14] proves that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold X . This note studies a finer count on the same generic even quantum D -module. It retains the marked small section and the original loop framing and counts primitive-sixth formal-monodromy eigenvalues. We denote the resulting count by ν_6 .

The cubic calculation and the product formula are unconditional:

$$\nu_6(X) = 2, \quad \nu_6(X \times \mathbf{P}^1) = 4.$$

The operation formulas require one additional reconstruction statement, and the remaining low-dimensional center argument requires a separate divisor-tagging statement for surface centers that are neither minimal nor geometrically ruled. We state both precisely where they enter. A strictly admissible F_1 specialization has $\nu_6 = 2$, whereas independent divisor tags give zero (Proposition 2.18). Thus universal tagging invariance fails; the remaining hypothesis is restricted to actual center comparison maps.

Theorem 1.1 (Conditional framed obstruction). *Assume Hypothesis R below, and Hypothesis T only for surface centers that are neither minimal nor geometrically ruled. For every smooth complex cubic threefold X ,*

$$\nu_6(X \times \mathbf{P}^1) = 4,$$

every rational smooth projective fourfold has $\nu_6 = 0$, and consequently $X \times \mathbf{P}^1$ is irrational.

The conclusion is not needed for the unconditional theorem of [14]. Its purpose is to isolate a finer invariant and the exact geometric comparison inputs still needed to make its birational invariance unconditional.

Section 2 defines the count, proves the cubic and product formulas, then derives the conditional operation laws, center vanishing, and the theorem.

2 Framed formal monodromy: a conditional refinement

The direct-QDM paper [14] proves an occurrence-indexed marker-ledger theorem and applies its rank-two formal-exponent fold. We keep that ledger construction and target monoid $A = \mathbf{N}$ fixed, refine the block type by a marked small section and loop framing, and replace the fold by primitive-sixth framed formal-monodromy multiplicity. The argument is a second specialization of the same marker theorem, not a parallel weak-factorization proof.

The Iritani and Iritani–Koto comparison isomorphisms preserve the original loop framing after coefficient-field extension. The string and divisor equations remove the scalar degree-zero term and every fixed degree-two term whose Novikov character descends to the coefficient ring. The additional inputs needed to make the operation formulas compatible with this finer fold are a residual reconstruction-tail displacement and a logically separate specialization step in the divisor-tagging argument. These are Hypotheses R and T.

The framed conclusion is logically subordinate to the unconditional theorem of [14]. Definition 2.1, the cubic computation, and the product-with-projective-space formula below are unconditional. The projective-bundle and blowup operation formulas use Hypothesis R. We prove specialized low-dimensional vanishing directly, without Hypothesis T, for nef-canonical targets, for $\mathbf{P}^1$ and $\mathbf{P}^2$, for every rational geometrically ruled surface—for F_1 on specializations that are graded-monomial in the sense of Definition 2.11, as every center specialization is—and, assuming only Hypothesis R, for ruled surfaces over curves of positive genus. In the present proof Hypothesis T is used only for surface centers that are neither minimal nor geometrically ruled. Consequently the resulting birational invariance of ν_6 , and the identification $\nu_6(V) = 2$ for genus-eight Fano threefolds, remain conditional on Hypothesis R and on Hypothesis T for those centers.

The exponent interpretation in [14, Section 2] explains how the two folds are related on the cubic rank-two block. The direct fold tests whether the two formal exponent classes are distinct modulo $\mathbf{Z}$. The auxiliary quantity $\delta^\sharp$ is the square of the difference of the residue representatives selected by the canonical elementary modification; for the cubic it equals $4/9$. The framed invariant ν_6 is finer: it tests the absolute cyclotomic placement of the formal exponent classes, together with descent data for the original loop framing when ramification is present. For the cubic zero block, [14, Section 2] finds the representatives $-1/6$ and $-5/6$; the calculation below recovers those same exponents directly from formal solutions and records that their classes give the two primitive sixth roots. Thus The direct marker and the framed count use two levels of the same formal data rather than unrelated numerical obstructions.

Let T be a smooth projective variety. Write

$$\Lambda_T = \mathbf{C}[[\mathrm{NE}_{\mathrm{num}}^{\mathbf{Z}}(T)]]$$

for the completed monoid ring of the image of effective homology classes in the numerical curve lattice

$$N_1(T) = \mathrm{im}(H_2(T, \mathbf{Z}) \longrightarrow \mathrm{Hom}(\mathrm{NS}(T)/\mathrm{tors}, \mathbf{Z})).$$

Thus $N_1(T)$ is torsion-free and is separated by integral divisors. The completed monoid ring Λ_T is a domain. Indeed, the effective numerical monoid embeds in the torsion-free lattice $N_1(T)$, hence is cancellative and its ordinary monoid algebra is a domain. An ample degree is positive and proper on the effective numerical monoid, so every nonzero completed series has a lowest nonzero homogeneous part, with only finitely many monomials in that degree. The lowest homogeneous part of a product is the product of the two lowest homogeneous parts and is nonzero in the monoid algebra. Therefore $\mathrm{Frac}(\Lambda_T)$ is defined. Let $\mathcal{K}_T$ be an algebraic closure of this fraction field. The loop coordinate z remains independent: in

particular, $\mathcal{K}_T$ contains no chosen root of z . We first isolate the cyclotomic part of formal monodromy without choosing values for the exponentials of arbitrary elements of $\mathcal{K}_T$.

Let $K \supset \mathbf{C}$ be an algebraically closed field of characteristic zero. Choose a complement $K = \mathbf{C} \oplus V$ of $\mathbf{Q}$ -vector spaces, let $K[V]$ be the group algebra of the additive group of V , and let Ω_V be an algebraic closure of its fraction field. Thus $K \subset \Omega_V$. The map

$$\text{Exp}_V(c + v) = e^{2\pi i c}[v]: (K, +) \longrightarrow \Omega_V^\times$$

is a group homomorphism. Since V is torsion-free, $K[V]$ is a domain, and its distinct monomials are linearly independent. Consequently

$$\ker(\text{Exp}_V) = \mathbf{Z}, \quad \text{Exp}_V(a) \text{ is a root of unity} \iff a \in \mathbf{Q}.$$

Apply Levelt–Turrittin after a ramification $z = w^e$. In the formal decomposition, write the residue on a regular-singular block as $R = R_s + R_n$, with commuting semisimple and nilpotent parts. In the universal formal solution algebra, define a full turn of w on that block by

$$M_{\text{RS},V} = \text{Exp}_V(R_s) \exp(2\pi i R_n),$$

where Exp_V is applied to the eigenvalues of R_s . The lift of one turn of the original z -disc acts on the formal symbols by $w \mapsto e^{2\pi i/e}w$, and on the blocks by their descent intertwiners. Its e -th power is $M_{\text{RS},V}$ on every returned regular-singular block. An integral change of residue representative multiplies the formal solution by an integral power of w , so it conjugates this lifted action rather than changing it. Since the coefficient and descent matrices lie in $K \subset \Omega_V$, this defines the framed operator $M_{f,V}$ over Ω_V . Ramified exponential factors are thereby permuted rather than made single-valued by changing the frame.

Choice independence. For every root of unity $\zeta \in \mathbf{C}$, its algebraic multiplicity as an eigenvalue of $M_{f,V}$ is independent of the complement V , the Levelt–Turrittin ramification and decomposition, and the algebraic closure Ω_V . It is also unchanged when the chosen algebraic closure of the original coefficient field is replaced by a common algebraically closed overfield. When $K = \mathbf{C}$, it is the usual multiplicity in formal monodromy on the original z -disc. This is distinct from full analytic monodromy of an irregular connection, whose comparison of sectorial solutions also involves Stokes matrices [12, Lecture 5]. We use here the functoriality and uniqueness of the ramified formal decomposition. That statement holds over any algebraically closed coefficient field of characteristic zero, which is the algebraic form we need, and in that generality it is the formal classification of differential modules over $K((z))$ [11, Chapter 3]. Sabbah gives the complex-analytic account, where the same decomposition appears alongside the Riemann–Hilbert correspondence [12, Lecture 5, Section 5.c, proofs of Theorem 5.8 and Proposition 5.10]. Consider one orbit of Levelt–Turrittin blocks under a turn of the original disc, let d be its length, and write $A_i: E_i \longrightarrow E_{i+1}$ for its successive deck/descent maps, with indices modulo d . Put $U = A_{d-1} \cdots A_0$ on E_0 . A cyclic-block determinant gives

$$\chi_{\bigoplus_i E_i}(X) = \det(X^d I - U).$$

Put $r = e/d$. Equivariance of the descent datum gives the intrinsic return relation

$$U^r = M_{\text{RS},V}.$$

Thus, if μ is an eigenvalue of U , then $\mu^r = \text{Exp}_V(\rho)$ for a residue exponent ρ , with algebraic multiplicities read on the corresponding generalized eigenspaces. If a root of unity ζ occurs in (2), then $\mu = \zeta^d$ and $\text{Exp}_V(\rho) = \zeta^e$; hence $\rho \in \mathbf{Q}$ by (2). Conversely, if ρ is rational, then

μ^r is torsion, so every corresponding μ , and hence every root of $X^d - \mu$, is torsion. On these rational residue classes Exp_V is the usual complex exponential; the finite descent operator is the same intrinsic operator. The exact roots of unity and their algebraic multiplicities are therefore those of the ordinary complex construction. Nonrational residue classes contribute no root of unity. This criterion does not depend on V and does not require a fractional residue factor to commute with the descent action.

Compare two Levelt–Turrittin choices after pulling both to a common ramification $z = \tilde{w}^E$. Equivariant uniqueness of the formal decomposition identifies their total horizontal solution spaces and carries the original- z -turn deck/descent operator for one choice to the scalar extension of the other. Thus the product of all orbit polynomials (2), rather than any individually labelled return map, is invariant under refinement and re-decomposition. The same argument after a common algebraically closed coefficient extension preserves the intrinsic multiset of residue classes and finite descent characters. The relation (2) is itself invariant under integral changes of residue representative. Hence the cyclotomic test above and its algebraic multiplicities depend only on the original framed module, not on the coefficient closure or complement. Finally, for $K = \mathbf{C}$ one has $V = 0$ and $\text{Exp}_V(a) = e^{2\pi i a}$, which gives the usual analytic formal monodromy.

Definition 2.1. Apply the preceding construction to the small even quantum connection over $\mathcal{K}_T((z))$, and put

$$\nu_6(T) = \text{mult}_{e^{\pi i/3}} M_{f,V}(T) + \text{mult}_{e^{-\pi i/3}} M_{f,V}(T).$$

The multiplicities are algebraic multiplicities in the characteristic polynomial. The choice-independence argument above makes their sum independent of all the auxiliary choices.

Let $\mathcal{T}_{\text{fr}}$ be the refinement of the generic-even block type whose objects retain the chosen small section and original z -disc framing, and whose morphisms preserve both. Let $\Pi_{\mathcal{T}_{\text{fr}}}$ and $\mathcal{L}_{\mathcal{T}_{\text{fr}}}$ be its groupoid and effective block monoid. For a block E of this type, let $w_6(E)$ be the same sum of the two primitive-sixth algebraic multiplicities, and extend blockwise:

$$w_6 : \mathcal{L}_{\mathcal{T}_{\text{fr}}} \longrightarrow \mathbf{N}, \quad \nu_6(T) = w_6(\mathcal{B}_{\mathcal{T}_{\text{fr}}}(T)).$$

Direct-sum additivity of characteristic polynomials makes w_6 a commutative-monoid fold. Lemma 2.2 shows that it is preserved by the regular scalar extensions and formal-coordinate isomorphisms in the common QDM construction. Neither marked datum is forgotten when passing to $\mathcal{L}_{\mathcal{T}_{\text{fr}}}$.

The frame matters. Adjoining roots of Novikov monomials is an algebraic coefficient extension and is harmless by choice independence. Adjoining a root of z , by contrast, would replace the original loop and could erase the deck part of the operator.

Lemma 2.2 (Frame transport over a coefficient field). *Let $K \supset \mathbf{C}$ be an algebraically closed field of characteristic zero, and let D, D' be differential modules over $K((z))$. Suppose $G \in \text{GL}_n(K((z)))$ is a gauge isomorphism from D to D' . Then the framed operators $M_{f,V}(D)$ and $M_{f,V}(D')$, defined using the original z -disc turn, are conjugate after passage to a common coefficient extension. In particular the algebraic multiplicity of every root of unity is the same for D and D' .*

Proof. After a common Levelt–Turrittin ramification, choose a formal solution matrix Y for D in the formal solution algebra used in Definition 2.1, and let σ denote one turn of the original z -disc. Since G has entries in $K((z))$, $\sigma(G) = G$. Hence

$$\sigma(GY) = G\sigma(Y).$$

Thus GY is a solution frame for D' in which the original-disc turn has the same matrix as in the frame Y . Comparing with any other solution frame for D' conjugates that matrix by a constant matrix. Therefore the framed operators are conjugate and have the same characteristic polynomial. $\square$

Lemma 2.3 (Exact string and fixed-divisor shifts). *Let a pullback of the even quantum connection of a smooth projective variety T be defined over a coefficient field containing its Novikov coefficients.*

- (i) *Replacing a bulk point τ by $\tau + c\mathbf{1}$ changes the z -connection only by the scalar irregular twist $e^{-c/z}$. Hence it does not change the framed formal-monodromy eigenvalues on the original z -disc, for any coefficient c for which the shifted connection is defined.*
- (ii) *Let $D \in H^2(T) \otimes K$ be a bulk shift such that $d \mapsto e^{\langle D, d \rangle}$ defines a multiplicative character on the numerical Novikov image in the coefficient field. Then the connection at $\tau + D$ is obtained from that at τ by the coefficient substitution*

$$Q^d \mapsto e^{\langle D, d \rangle} Q^d.$$

This substitution fixes $\mathbf{C}$, so it preserves the algebraic multiplicities of every root of unity in framed formal monodromy.

Proof. Write the quantum connection in the convention of [7, (2.2)–(2.3) and Remark 2.3]. The String Equation implies that the quantum product is independent of the H^0 coordinate, while the Euler field acquires $c\mathbf{1}$. Thus the horizontal equation $z^2 \partial_z Y = (U - z\mu)Y$ acquires the scalar term cI , and multiplication of a solution by $e^{-c/z}$ gives the shifted equation. This exponential is single-valued under a full turn of the original z -disc and changes no regular exponent or descent datum.

For a degree-two shift, the Divisor Equation says that the quantum product depends on the divisor coordinate only through the combinations $Q^d e^{\langle D, d \rangle}$; the degree-two coordinate does not occur in the Euler field. Therefore the full z -connection is the coefficient base change displayed above. The induced embedding of coefficient fields fixes the complex roots of unity, and choice independence in Definition 2.1 gives the last assertion. $\square$

Remark 2.4 (Why a residual hypothesis remains). Flatness does give a normalized gauge for a genuinely formal bulk germ. Writing the formal bulk variables collectively as η and solving

$$\partial_{\eta_i} G = -z^{-1}(\phi_i \star_{\eta}) G, \quad G|_{\eta=0} = I,$$

recursively, the coefficient of total bulk degree m has no power of z below $-m$. Thus every fixed bulk degree uses only integral powers of the original loop coordinate and is harmless for framing. After substituting a reconstruction coordinate containing Novikov coefficients, however, the sum over all bulk degrees need not have a uniform lower Laurent bound in z . We therefore do not promote this degreewise gauge to an element of a Laurent series field and do not use it to identify intrinsic framed monodromy at a reconstruction value. Hypotheses R and T below isolate exactly the two places where such an identification is still required. The rank-two theorem below avoids this gauge altogether on the formal germ where it applies.

Lemma 2.5 (Numerical Novikov base change). *The quotient from the effective homology monoid to its image in $N_1(T)$ extends continuously to completed monoid rings, and the quantum connection and the two comparison theorems base-change along it.*

Proof. Filter both monoids by intersection with an ample divisor. Effective homology classes below a fixed cutoff form a finite set, as follows from the finite-type Chow varieties of cycles of bounded degree. The quotient is therefore defined on every truncated monoid ring and passes to the inverse limit. Explicitly, the coefficient of $Q^{\bar{d}}$ in the image of $\sum_d a_d Q^d$ is

$$\sum_{d \rightarrow \bar{d}} a_d.$$

The sum is finite at every cutoff, and additivity of the quotient makes this coefficient-wise pushforward a unital homomorphism for completed convolution. Thus the quantum products descend by grouping Gromov–Witten coefficients with the same numerical class. Pushforward of curves, pairing with $c_1(N_{C/T})$, and the derivations $Q\partial_Q$ are constant on these fibres. At each cutoff, apply the resulting ring homomorphism to the comparison map and to its stated inverse. Scalar extension preserves their two inverse identities. Grouping coefficients identifies the scalar-extended quantum module with the numerical quantum module, and the preceding fiberwise constancy identifies its connection operators and divisor substitutions. Passing through the compatible cutoffs proves the asserted completed base change. $\square$

Proposition 2.6 (Formal-germ rigidity of an isolated rank-two packet). *Let K be an algebraically closed coefficient field of characteristic zero and let the even quantum connection be pulled back to a formal smooth bulk germ $B = K[[t_1, \dots, t_m]]$. Suppose that at the closed point the Euler operator has an eigenvalue whose generalized eigenspace has rank two, its nilpotent part is a single nonzero Jordan block, and every other Euler eigenvalue differs from it by a unit of K . Then the corresponding rank-two spectral cluster remains a single nonzero Jordan block on the formal germ. After scalar centering and the canonical elementary modification of [14, Section 2], its residue R has constant characteristic polynomial on B . Consequently the algebraic multiplicities of the primitive sixth roots contributed by this block are constant on the formal germ.*

Proof. The unit separation from the other Euler eigenvalues gives a formal spectral projector onto the rank-two cluster. Put $U_0 = \lambda I + N$ on this cluster, with $\text{tr } N = 0$, and let C_a denote quantum multiplication by the bulk tangent vector ∂_{t_a} . Flatness gives the coefficient identities

$$[U, C_a] = 0, \quad \partial_{t_a} U = C_a + [C_a, \mu].$$

Let P be the formal spectral projector onto this cluster and use the induced connection D_a on $P(H)$. For an endomorphism preserving the cluster, D_a is the projected derivative, so

$$D_a U_0 = P(\partial_{t_a} U)P.$$

Because $[U, C_a] = 0$, the operator C_a commutes with P . Compressing the second flatness identity therefore gives

$$D_a U_0 = C_{a,0} + [C_{a,0}, \mu_0], \quad C_{a,0} = P C_a P, \quad \mu_0 = P \mu P.$$

Because the reduction of U_0 at the closed point is nonscalar, U_0 is a regular 2×2 matrix over the local ring B . After choosing the closed-point Jordan frame, one off-diagonal entry of N is a unit; solving $[N, X] = 0$ directly in that frame gives the commutant $BI \oplus BN$. Hence

$$C_{a,0} = p_a I + q_a N.$$

Taking traces in the block evolution gives $p_a = \partial_{t_a} \lambda$ and therefore

$$D_a N = q_a N + q_a [N, \mu_0].$$

Set $d = \det N$. Since determinant is invariant under change of frame, its ordinary derivative can be computed with the covariant derivative D_a . For a traceless 2×2 matrix, $\text{adj}(N) = -N$ and $N^2 = -dI$. Cyclicity of trace then gives

$$\partial_{t_a} d = \text{tr}(\text{adj}(N)D_a N) = 2q_a d.$$

Since d vanishes at the closed point, a lowest-homogeneous-term argument in $K[[t_1, \dots, t_m]]$ forces $d = 0$. Thus $N^2 = 0$ everywhere. A matrix entry of N is a unit at the closed point, so N remains nonzero and its image is a line subbundle.

The rank-two rigidity proof in [14, Section 2] is purely algebraic: it uses only flatness, Frobenius self-adjointness, anti-self-adjointness of the grading operator, and the nondegenerate Poincaré pairing on the separated spectral factor. Those identities hold coefficientwise over the present formal germ, so the same elementary modification is regular in every base direction and its residue satisfies

$$D_a R = [G_a, R].$$

Hence $\text{tr } R$ and $\det R$, and therefore the characteristic polynomial of R , are constant. By the exponent interpretation there, the residue eigenvalues are representatives of the formal exponent classes; the integral shear changes them only by integers. The eigenvalues of the regular formal monodromy are therefore constant, including in resonant cases where only the unipotent part could vary. This proves the cyclotomic multiplicity statement. $\square$

Remark 2.7 (Formal-germ scope). Proposition 2.6 is deliberately a theorem over a formal bulk germ with a fixed coefficient field. It does not identify an Iritani or Iritani–Koto reconstruction value with a point of that germ. The comparison theorems are formulated over graded Laurent completions in Novikov variables, whereas the proposition deforms genuinely formal bulk variables over K . Substituting a Laurent–Novikov reconstruction expression for those formal variables is an additional operation, and no theorem legitimizing that substitution for framed monodromy is assumed here. Thus the proposition proves the local rank-two mechanism behind Hypothesis R; it is not a proof of that hypothesis at a reconstruction value.

Hypothesis R (Reconstruction-tail invariance). *Consider only the residual reconstruction displacements that occur in the blowup and projective-bundle decompositions used in Proposition 2.8: the $O(u) + s_j$ tails on the target summands and the corresponding residual ambient shift, after the scalar H^0 term and every fixed H^2 term whose Novikov character descends to the relevant coefficient ring have been removed by Lemma 2.3. After the coefficient specialization displayed in the construction, replacing zero bulk by one of these residual reconstruction displacements does not change the algebraic multiplicities of $e^{\pi i/3}$ and $e^{-\pi i/3}$ in framed formal monodromy on the original z -disc.*

No invariance under an arbitrary bulk displacement is asserted.

Proposition 2.8 (Framed operation formulas, conditional). *Assume Hypothesis R. Let $V \rightarrow T$ be a vector bundle of rank $r \geq 2$. Then*

$$\nu_6(\mathbf{P}_T(V)) = r\nu_6(T).$$

Let $\tilde{T} = \text{Bl}_C T$, where $C \subset T$ is smooth of codimension $c \geq 2$. In Iritani’s comparison field one has

$$\nu_6(\tilde{T}) = \nu_6(T) + \sum_{j=0}^{c-2} \nu_6(C; \chi_j),$$

where χ_j is the numerical Novikov specialization of the j -th center summand. The endpoint terms in (2.8) are the intrinsic small invariants; the center terms are computed after the displayed specializations.

Proof. The QDM operation theorem of [14, Section 2] supplies the common direct-sum, coefficient-spine, parity, and occurrence bookkeeping. Lemma 2.2 shows that the fold w_6 crosses its regular comparison maps without changing the original z -disc. It remains only to compare the reconstructed bulk points with the intrinsic small sections.

For a blowup center summand put $u = q^{-1/(c-1)}$. Iritani's coordinates have the initial form

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = -(c-1)\lambda_j + h_{C,j} + O(u),$$

with $\lambda_j = \zeta_j u^{-1}$ [7, Section 5.8.2 and (5.27)–(5.30)], with numerical specialization

$$\chi_j(Q_C^d) = Q^{i_*d} u^{\rho_C \cdot d}. \quad (2.1)$$

Lemma 2.3 removes the scalar term and the fixed divisor $h_{C,j}$ exactly. The latter character descends through (2.1) because $h_{C,j}$ is a scalar multiple of ρ_C . Section 5.8.1 of [7] fixes the logarithmic normalization. Hypothesis R is invoked exactly for the remaining $O(u) + s_j$ tail, and analogously for the residual ambient tail $\tau^\circ = O(q^{-1})$. Additivity of w_6 now gives (2.8). The center term retains χ_j , since the map (2.1) need not be injective.

For a projective bundle put $u = q^{-1/r}$. Iritani–Koto's reconstruction coordinates satisfy

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = r\lambda_j - \frac{2\pi i j}{r} c_1(V) + O(u)$$

[8, Theorem 5.1(4), Proposition 5.6]. The exact shift lemma removes the first two terms, while [8, (5.11)–(5.12)] fixes the initial coordinate. Hypothesis R is used only for the residual $O(u) + s_j$ tail. The base Novikov map is injective [8, (1.1) and Remark 5.2], so each of the r ledger blocks contributes $\nu_6(T)$. This proves (2.8). $\square$

The center map in (2.8) is not necessarily injective [7, (5.15)]. Thus generic vanishing for C does not by itself imply that its specialized term vanishes. The next lemma supplies the missing implication.

Definition 2.9. A monomial map $\chi : \Lambda_T \rightarrow A$ is *strictly Novikov-admissible* if A is a complete separated valued $\mathbf{C}$ -domain with domain associated graded, every monomial has nonzero image, and

$$v(\chi(Q^d)) = \ell(d),$$

where ℓ is positive and proper on the effective numerical monoid.

We write $\nu_6(T; \chi)$ for the framed small-even invariant after scalar extension along such a map.

The center maps in (2.8) are strictly admissible. Give the exceptional Laurent variable valuation zero and use an ample divisor H on T . Then

$$v_H(Q^{i_*d} q^{-\rho_C \cdot d/(c-1)}) = (H|_C) \cdot d.$$

The restriction $H|_C$ is ample, which gives positivity and properness; the associated graded Laurent monoid ring is a domain.

Proposition 2.10 (Direct specialized low-dimensional vanishing). *Let $\chi : \Lambda_T \rightarrow A$ be strictly Novikov-admissible.*

(i) *If the canonical divisor K_T is nef, then*

$$\nu_6(T; \chi) = 0$$

unconditionally.

(ii) If $T = \mathbf{P}^1$ or $T = \mathbf{P}^2$, then

$$\nu_6(T; \chi) = 0$$

unconditionally.

(iii) Let $T = \mathbf{P}_C(V)$ be a geometrically ruled surface with $g(C) \geq 1$. Assuming Hypothesis R,

$$\nu_6(T; \chi) = 0.$$

Proof. For (i), work after scalar extension to an algebraic closure of $\text{Frac } A$. At small bulk write

$$\nabla_{z\partial_z} = z\partial_z - z^{-1}U + \text{Gr}, \quad U = \sum_{\beta} \chi(Q^\beta)U_\beta,$$

where U_β is the coefficient of Euler multiplication in numerical class β . Put

$$g = \text{Gr} + \frac{1}{2}E,$$

with E equal to the identity on cohomological degrees of parity opposite to $\dim T$ and zero on the other parity. The operator g has integral eigenvalues. The genus-zero dimension constraint gives

$$U_\beta : H^i(T) \longrightarrow H^{i+2-2c_1(T)\cdot\beta}(T),$$

so, because quantum multiplication preserves parity, U_β raises g -weight by

$$m_\beta = 1 - c_1(T) \cdot \beta.$$

If K_T is nef, then $m_\beta \geq 1$ for every effective β . Make the half-parity correction explicitly by replacing the z -connection with

$$\nabla_{z\partial_z}^{\text{par}} = \nabla_{z\partial_z} + \frac{1}{2}E = z\partial_z - z^{-1}U + g.$$

With the gauge convention $G\nabla G^{-1}$, take $G = z^g$. Since $[g, U_\beta] = m_\beta U_\beta$, one has

$$z^g U_\beta z^{-g} = z^{m_\beta} U_\beta, \quad z^g (z\partial_z) z^{-g} = z\partial_z - g,$$

and therefore

$$z^g \nabla_{z\partial_z}^{\text{par}} z^{-g} = z\partial_z - \sum_{\beta} \chi(Q^\beta) z^{m_\beta-1} U_\beta.$$

Thus the specialized connection is regular singular. This conclusion is unchanged when the noninjective map χ groups several numerical classes: every grouped summand still has nonnegative z -order. The residue consists only of the terms with $m_\beta = 1$, equivalently $c_1(T) \cdot \beta = 0$. Each such operator raises the finite g -filtration by one, so their sum is nilpotent. The parity-corrected regular monodromy is therefore unipotent. Undoing the half-parity correction permits only eigenvalues 1 and -1 , because the transformed residue preserves parity. Hence no primitive sixth root occurs. Strict Novikov-admissibility is used here only to ensure that the specialized small connection is a well-defined coefficientwise specialization; injectivity is not used.

For (ii), let ℓ be the line class and put $a = \chi(Q^\ell)$. Strict admissibility gives $a \neq 0$. Over an algebraic closure of $\text{Frac } A$, the specialized small quantum relations are $H^2 = a$ for $\mathbf{P}^1$ and $H^3 = a$ for $\mathbf{P}^2$. Thus Euler multiplication has respectively two and three distinct eigenvalues. Every spectral block has rank one, so Lemma 2.22 gives trivial framed regular monodromy on every block.

For (iii), let f be the fibre class. Every map from a connected nodal genus-zero curve to C is constant: it is constant on every irreducible rational component, and connectedness forces the same image point on all components. Hence every nonzero curve class contributing to the genus-zero quantum product of T is a positive multiple of f . The small quantum differential module is therefore obtained by scalar extension from a one-variable module over $\mathbf{C}[[Q^f]]$. The restriction of χ to the fibre ray is injective, because

$$v(\chi(Q^{nf})) = n\ell(f), \quad \ell(f) > 0,$$

so distinct powers have distinct valuation. On the other hand, Hypothesis R and Proposition 2.8 give the intrinsic projective-bundle identity

$$\nu_6(T) = 2\nu_6(C) = 0,$$

where the last equality follows by applying (i) to the identity Novikov specialization of C , since K_C is nef for $g(C) \geq 1$. The intrinsic small module and the χ -specialized module are both scalar extensions of this same one-variable differential module. More explicitly, let

$$K_f = \text{Frac}(\mathbf{C}[[Q^f]]), \quad K_{f,\chi} = \text{Frac}(\chi(\mathbf{C}[[Q^f]])) \subset \text{Frac}(A).$$

The injective restriction of χ identifies K_f with $K_{f,\chi}$. Via this identification, choose a common algebraically closed overfield Ω containing K_f and $\text{Frac}(A)$. After scalar extension to Ω , the intrinsic one-variable module and its χ -specialization become literally the same differential module. Definition 2.1 and its choice-independence under common coefficient-field extension therefore give identical root-of-unity multiplicities. The specialized value also vanishes. $\square$

The remaining centers are the rational geometrically ruled surfaces, that is the Hirzebruch surfaces $F_a = \mathbf{P}^1(\mathcal{O} \oplus \mathcal{O}(a))$; these are the minimal surfaces not yet covered, together with F_1 , which is geometrically ruled without being minimal. For those the projective-bundle comparison is available only intrinsically, so we compute the specialized Euler spectrum instead. The quadric surface $F_0 = \mathbf{P}^1 \times \mathbf{P}^1$ is handled separately by the Gromov–Witten product formula, because χ may send its two ruling classes to the same value and the quartic below then degenerates. For $a \geq 1$ two facts carry the computation: one deformation carries the genus-zero theory of F_a to that of F_0 or F_1 , whose quantum relations serve as the base cases, under an explicit change of Novikov variables, and the resulting rank-four Euler quartic is separable away from one explicit locus that no center specialization meets.

One case, the surface F_1 , needs a mild extra condition on χ , used only to rule out a single algebraic coincidence in the associated graded ring. The word *monomial* in Definition 2.9 refers to the source, a map defined on Novikov monomials; the condition below is about the target, so we call it *graded-monomial*.

Definition 2.11. A strictly Novikov-admissible $\chi : \Lambda_T \rightarrow A$ is *graded-monomial* if the associated graded ring $\text{gr } A$ admits a $\mathbf{C}$ -basis B containing the leading term $\text{gr } \chi(Q^d)$ of $\chi(Q^d)$ for every effective class d .

Lemma 2.12 (Center specializations are graded-monomial). *The specializations (2.1) are graded-monomial.*

Proof. The leading term of $\chi_j(Q_C^d) = Q^{i_*d} u^{\rho_C \cdot d}$ is its coefficient-one Laurent monomial in the associated graded of A . Distinct such monomials are $\mathbf{C}$ -linearly independent, even though the valuation-zero variable u allows the degree-zero coefficients of the completion to be larger than $\mathbf{C}[u^{\pm 1}]$. Extend this linearly independent set to a $\mathbf{C}$ -basis of $\text{gr } A$. The resulting basis contains the leading term of every effective center monomial. $\square$

Lemma 2.13 (Specialized Euler spectrum of a Hirzebruch surface). *Let $T = F_a = \mathbf{P}_{\mathbf{P}^1}(\mathcal{O} \oplus \mathcal{O}(a))$ with $a \geq 0$, let f and s be the class of a fibre and the class of a section of self-intersection $-a$ in $N_1(T)$, and put $k = \lfloor a/2 \rfloor$. For $a = 0$ either ruling may be taken as the fibre; the quartic below is symmetric in u and w . Let $\chi : \Lambda_T \rightarrow A$ be strictly Novikov-admissible and set*

$$u = \chi(Q^f), \quad w = \chi(Q^{s+kf}).$$

After scalar extension to an algebraic closure of $\text{Frac } A$, the characteristic polynomial of Euler multiplication on $H^0(T) \oplus H^2(T) \oplus H^4(T)$ is

$$P_a(X) = \begin{cases} X^4 - 8(u+w)X^2 + 16(u-w)^2, & a \text{ even}, \\ X^4 + wX^3 - 8uX^2 - 36uwX + 16u^2 - 27uw^2, & a \text{ odd}, \end{cases}$$

and its discriminant is

$$\text{disc } P_a = \begin{cases} 2^{24} u^2 w^2 (u-w)^2, & a \text{ even}, \\ -u^2 w^2 (256u + 27w^2)^3, & a \text{ odd}. \end{cases}$$

Proof. Write F and S for the divisor classes corresponding to f and s , the class of a fibre and the class of the section of self-intersection $-a$, so that $F^2 = 0$, $S \cdot F = 1$, $S^2 = -a$ and $c_1(T) = 2S + (a+2)F$. Put $S^b = S + kF$, of self-intersection $-a + 2k \in \{0, -1\}$; then

$$c_1(T) = \begin{cases} 2S^b + 2F, & a = 2k, \\ 2S^b + 3F, & a = 2k + 1. \end{cases}$$

A single deformation to $a \leq 1$. Let $a \geq 2$. Since $k \geq 1$ and $a - k \geq 1$, a general pair of sections of $\mathcal{O}(k)$ and $\mathcal{O}(a-k)$ has no common zero, so it defines an injection $\mathcal{O} \rightarrow \mathcal{O}(k) \oplus \mathcal{O}(a-k)$ with locally free quotient, necessarily $\mathcal{O}(a)$ by degree. Let $\varepsilon \in \text{Ext}^1(\mathcal{O}(a), \mathcal{O})$ be the class of the resulting extension and consider the extensions E_t with class $t\varepsilon$, $t \in \mathbf{A}^1$, that is the pullback of the universal extension along $t \mapsto t\varepsilon$. These form a vector bundle on $\mathbf{P}^1 \times \mathbf{A}^1$. Scaling a nonzero extension class by a unit leaves the middle term unchanged, so $E_t \cong E_1$ for every $t \neq 0$. The extension is nonsplit, since $\mathcal{O}(k) \oplus \mathcal{O}(a-k)$ is not $\mathcal{O} \oplus \mathcal{O}(a)$, so $\varepsilon \neq 0$. The projectivization of the family is smooth and projective, with

$$\mathbf{P}(E_0) = \mathbf{P}(\mathcal{O} \oplus \mathcal{O}(a)) = F_a, \quad \mathbf{P}(E_t) = \mathbf{P}(\mathcal{O}(k) \oplus \mathcal{O}(a-k)) \cong F_{a-2k} \quad (t \neq 0),$$

the last isomorphism by twisting with $\mathcal{O}(-k)$. Genus-zero Gromov–Witten invariants are invariants of a deformation class [10, Chapter 7]; a smooth projective family carries a smooth family of Kähler forms, and every symplectic four-manifold is semipositive, so the genus-zero invariants of [10] are defined and deformation-invariant for this family. The invariants therefore agree under the parallel transport ψ of N_1 along the family. The ruling is global over the base, so ψ fixes f ; writing $\psi(s) = \alpha s_0 + \beta f$ with s_0 the negative-section class of F_{a-2k} , which for F_0 means the other ruling class, pairing with f gives $\alpha = 1$ and $\psi(s)^2 = -a$ gives $\beta = -k$. Thus

$$\psi(mf + ns) = (m - nk)f + ns_0, \quad \psi(f) = f, \quad \psi(s + kf) = s_0,$$

and correspondingly $S^b \mapsto S_0$, the negative section of $F_{a-2k} = F_{a \bmod 2}$ as a divisor class. For $a \leq 1$ take ψ to be the identity.

The two base cases. For $F_0 = \mathbf{P}^1 \times \mathbf{P}^1$ the Gromov–Witten product formula [3] gives $QH^*(F_0) = QH^*(\mathbf{P}^1) \otimes QH^*(\mathbf{P}^1)$, so

$$S_0 \star S_0 = Q^f, \quad F \star F = Q^{s_0}.$$

For F_1 , which is Fano, the toric quantum Stanley–Reisner presentation of [1, Section 5]—established for Fano toric manifolds by the mirror theorem [6, Theorem 0.1]—gives

$$S_0 \star S_0 + S_0 \star F = Q^f, \quad F \star F = Q^{s_0} S_0.$$

The same two relations can be read off directly from the genus-zero invariants of F_1 , which is how we use them; the toric presentation is recorded as a check. On F_1 , $c_1 \cdot (mf + ns_0) = 2m + n$, so a three-point invariant with two divisor insertions and $\beta \neq 0$ is nonzero only for $1 \leq 2m + n \leq 2$; the contributing classes are s_0 , with $\langle \rangle_{s_0} = 1$ because the negative section is the unique curve in its class, and f , with $\langle \text{pt} \rangle_f = 1$ because one fibre passes through a general point. The class $2s_0$ contributes nothing, since every stable map in it has image the negative section and a general point is not on that curve. The divisor axiom then gives $S_0 \star S_0 = -\text{pt} + Q^{s_0} S_0 + Q^f$, $S_0 \star F = \text{pt} - Q^{s_0} S_0$ and $F \star F = Q^{s_0} S_0$, which is the displayed presentation.

Truncation. The transported theory determines the small quantum product of F_a itself, and only finitely many Novikov monomials survive. By the virtual dimension count, and by the fundamental-class axiom at the lower end, a three-point invariant of F_a with two divisor insertions and $\beta \neq 0$ vanishes unless $1 \leq c_1(T) \cdot \beta \leq 2$; and by deformation invariance it vanishes unless $\psi(\beta)$ is effective on $F_{a \bmod 2}$, that is unless $m \geq nk$ and $n \geq 0$ for $\beta = mf + ns$. Now

$$c_1(T) \cdot \beta = 2m + (2 - a)n = \begin{cases} 2(m - nk) + 2n, & a = 2k, \\ 2(m - nk) + n, & a = 2k + 1, \end{cases}$$

so $c_1(T) \cdot \beta \leq 2$ forces $n \leq 1$ for a even and $n \leq 2$ for a odd, and then $m = nk$ except for $\beta = f$. The surviving classes are f and $s + kf$ for a even, and f , $s + kf$ and $2(s + kf)$ for a odd. In particular the product involves no infinite Novikov sum, and the two monomials Q^f and Q^{s+kf} generate every coefficient that occurs. The displayed relations therefore hold in the specialized ring, and the resulting surjection from the presented algebra, which is free of rank four, onto the specialized quantum cohomology, also free of rank four, is an isomorphism.

The two transported presentations are those of [5, Theorems 9.3.1 and 9.3.3], obtained there by the same identification of the genus-zero theory of F_a with that of $F_{a \bmod 2}$ and, in that paper’s generators, by a shift of the section class by a multiple of the fibre class. What the quartic consumes, however, is the presentation of the *specialized* ring over A , and it is the truncation above that makes the transported product finite and identifies the presented algebra with that ring.

The quartic. Specializing along χ and substituting $Q^f \mapsto u$ and $Q^{s+kf} \mapsto w$ in the transported presentation, the specialized small quantum cohomology of F_a is $A[S^b, F]$ modulo $(S^{b^2} - u, F^2 - w)$ for a even and modulo $(S^{b^2} + S^b F - u, F^2 - wS^b)$ for a odd. In the basis $1, F, S^b, S^b F$ the matrix of multiplication by $c_1(T)$ is

$$\begin{pmatrix} 0 & 2w & 2u & 0 \\ 2 & 0 & 0 & 2u \\ 2 & 0 & 0 & 2w \\ 0 & 2 & 2 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 0 & 2u & wu \\ 3 & 0 & 0 & 2u \\ 2 & 3w & 0 & 0 \\ 0 & 2 & 1 & -w \end{pmatrix}$$

respectively, and its characteristic polynomial is (2.13). Evaluating the discriminant of a monic quartic on those coefficients gives (2.13). $\square$

Lemma 2.14 (Degeneracy dichotomy of the ruled Euler spectrum). *Keep the notation of Lemma 2.13; strict admissibility makes u and w nonzero. Exactly one of the following holds.*

(a) $\text{disc } P_a \neq 0$. Equivalently $u \neq w$ for a even and $256u + 27w^2 \neq 0$ for a odd. Then every spectral block of Euler multiplication has rank one.

(b) $\text{disc } P_a = 0$. Then P_a has exactly one double root and two simple roots, so Euler multiplication has one spectral block of rank two, whose nilpotent part squares to zero, and two blocks of rank one.

No other block shape occurs. By Lemmas 2.12 and 2.15, case (b) does not arise for a center specialization when $a \geq 1$; for F_0 it can, which is why that surface is treated by the product formula in Proposition 2.16.

Proof. By (2.13) and $u, w \neq 0$, the discriminant vanishes exactly on the stated locus. Off it the quartic is separable, so each of its four roots is simple and each maximal generalized eigenspace of Euler multiplication is a line.

On the locus the quartic factors. For a even, $u = w$ gives $P_a = X^2(X^2 - 16u)$, whose double root is 0 and whose two simple roots are the square roots of $16u$, distinct from 0 and from each other. For a odd, write $w = 16\sigma$; then $256u + 27w^2 = 0$ reads $u = -27\sigma^2$ and

$$P_a = (X + 18\sigma)^2 (X^2 - 20\sigma X + 612\sigma^2),$$

whose quadratic factor has the two roots $10\sigma \pm 16\varepsilon$ with $\varepsilon^2 = -2\sigma^2$; these are distinct from each other and from -18σ : a coincidence $10\sigma \pm 16\varepsilon = -18\sigma$ gives $4\varepsilon = \mp 7\sigma$, hence $49\sigma^2 = 16\varepsilon^2 = -32\sigma^2$ and $81\sigma^2 = 0$. In both cases the multiplicity of every root is at most two and the repeated root has multiplicity exactly two. A block of rank two has a single eigenvalue λ , hence trace 2λ and determinant λ^2 ; by Cayley–Hamilton its centered matrix $M - \lambda$ squares to zero. $\square$

Lemma 2.15 (Strictly admissible specializations avoid the degeneracy locus for $a \geq 1$). *Let $T = F_a$ with $a \geq 1$ and let $\chi : \Lambda_T \rightarrow A$ be strictly Novikov-admissible, and graded-monomial if $a = 1$. Then $\text{disc } P_a \neq 0$.*

Proof. Write $p = \ell(f)$ and $r = \ell(s)$; both are positive. Multiplicativity of the valuation and (2.9) give $v(u) = p$ and $v(w) = r + kp$.

Let $a = 2k$ with $k \geq 1$. Then $v(w) = r + kp > p = v(u)$, so $u \neq w$.

Let $a = 2k + 1$ with $k \geq 1$. Then $v(w^2) = 2r + 2kp > p = v(u)$, and a vanishing $256u + 27w^2$ would make $256u = -27w^2$. The integers 256 and 27 are units of A , and negation does not change a valuation, so this forces $v(u) = v(w^2)$, a contradiction.

Let $a = 1$, so $k = 0$ and $v(u) = p$, $v(w^2) = 2r$. If $p \neq 2r$ the previous argument applies. If $p = 2r$, work in the degree- p symbol group. Write $\sigma_p : A_{\geq p} \rightarrow A_{\geq p}/A_{>p}$ for the symbol map, which is additive. Both $256u$ and $27w^2$ have valuation exactly p , so

$$\sigma_p(256u + 27w^2) = 256 \sigma_p(u) + 27 \sigma_p(w^2),$$

and $\sigma_p(u) = \text{gr } \chi(Q^f)$ and $\sigma_p(w^2) = \text{gr } \chi(Q^{2s})$, the second because χ is a monoid map, so $w^2 = \chi(Q^{2s})$. Both are members of the basis B of Definition 2.11, since f and $2s$ are effective. Distinct basis members have no vanishing combination with nonzero coefficients, and equal ones would require $256 + 27 = 0$, so the symbol is nonzero and therefore $256u + 27w^2 \neq 0$. $\square$

Proposition 2.16 (Direct specialized vanishing for rational geometrically ruled centers). *Let $T = F_a$ with $a \geq 0$ and let $\chi : \Lambda_T \rightarrow A$ be strictly Novikov-admissible, and graded-monomial if $a = 1$. Then*

$$\nu_6(T; \chi) = 0.$$

Proof. Let $a = 0$, so $T = \mathbf{P}^1 \times \mathbf{P}^1$; this case is the χ -specialized analogue of Proposition 2.23 below. The Gromov–Witten product formula [3] identifies the small quantum differential module of a product with the tensor product of those of its factors over $\Lambda_{\mathbf{P}^1} \otimes \Lambda_{\mathbf{P}^1} = \Lambda_T$. Every entry of the resulting connection matrix lies in Λ_T , so applying χ entrywise commutes with the identification: the χ -specialized module of T is the tensor product of the modules specialized along the two restrictions of χ to the ruling classes f_1, f_2 . Each factor is a projective line with specialized quantum relation $H^2 = \chi(Q^{f_i})$, and strict admissibility keeps that value nonzero, so by the argument of Proposition 2.10(ii) each factor has simple Euler spectrum and, by Lemma 2.22, trivial framed regular monodromy. As in the proof of Proposition 2.23 below, the Levelt–Turrittin formal decomposition is compatible with tensor products of differential modules, so the framed operator of T is the tensor product of two identities. The argument uses no relation between the two specialized values, which is what the product route buys: χ may identify the two ruling classes, and the discriminant of Lemma 2.13 then vanishes.

Let $a \geq 1$. Lemma 2.15 gives $\text{disc } P_a \neq 0$, so by Lemma 2.14(a) every spectral block of Euler multiplication has rank one, and Lemma 2.22 makes the framed regular monodromy trivial on each. Hence no primitive sixth root occurs. $\square$

Remark 2.17 (Where the tagging hypothesis is still used). Proposition 2.10 shows that Hypothesis T is not needed for nef-canonical centers, for $\mathbf{P}^1$ or $\mathbf{P}^2$, or—assuming only Hypothesis R—for ruled surfaces over positive-genus curves, and Proposition 2.16 removes it for the rational geometrically ruled centers as well. Those cases exhaust the minimal surfaces, and also cover F_1 , which is geometrically ruled without being minimal. In the present proof of four-dimensional birational invariance, the substantive use of Hypothesis T is therefore confined to surface centers that are neither minimal nor geometrically ruled. Every such surface is an iterated point blowup, so eliminating it there would require a separate support or base-change statement for applying the blowup comparison after the external center specialization; no such statement is assumed here. The quartic of Lemma 2.13 was derived for the Hirzebruch surfaces and is not available for such a center.

Hypothesis T (Divisor-tagging specialization invariance). *Let $i : T \hookrightarrow Y$ be an actual smooth blowup center in a smooth projective variety of dimension at most four, with T a surface that is neither minimal nor geometrically ruled. Consider precisely its maps*

$$\chi_j(Q_T^d) = Q^{i_* d} u^{\rho_T \cdot d}, \quad u = q^{-1/(c-1)},$$

from (2.1), in the coefficient completion of that comparison, where $c = \text{codim}(T, Y)$. Allow the faithful coefficient-field extensions used there, but no further arbitrary specializations of the Novikov or exceptional variables. Write χ for one of these strictly Novikov-admissible maps. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(T)$ and define

$$\chi_t(Q^d) = \chi(Q^d) \exp \left(\sum_i t_i (D_i \cdot d) \right).$$

After passing to algebraic closures of the relevant fraction fields, the intrinsic framed operator of the χ -specialized small connection and the tagged module have the same algebraic multiplicities at $e^{\pi i/3}$ and $e^{-\pi i/3}$.

This hypothesis concerns only the specialization step in this tagging family. It does not assert reconstruction invariance or invariance under an arbitrary divisor-valued bulk deformation. Graded monomiality of these maps is proved in Lemma 2.12; it does not by itself prove this hypothesis for arbitrary surfaces.

Proposition 2.18 (Failure of unrestricted tagging invariance on F_1). *For the fibre f and negative section s of F_1 , the map*

$$\chi(Q^f) = -27t^2, \quad \chi(Q^s) = 16t \quad \text{in } \mathbf{C}[[t]]$$

is strictly Novikov-admissible but not graded-monomial. Its repeated Euler eigenvalue has a scalar rank-two leading block. The framed monodromy satisfies

$$\det(\lambda I - M_f) = (\lambda - 1)^2(\lambda^2 - \lambda + 1), \quad \nu_6(F_1; \chi) = 2. \quad (2.2)$$

Independent divisor tags $Q^f = -27t^2e^\alpha$, $Q^s = 16te^\beta$ give simple Euler spectrum and $\nu_6 = 0$. Thus Hypothesis T with unrestricted quantification over strictly admissible maps is false.

Proof. The valuation of the image of Q^{mf+ns} is $2m + n$, positive and proper on $\mathbf{N}^2$. Every monomial image is nonzero and $\text{gr } \mathbf{C}[[t]] = \mathbf{C}[t]$ is a domain. The initial forms $-27t^2$ and $256t^2$ of Q^f and Q^{2s} are distinct and proportional, so cannot both belong to a $\mathbf{C}$ -basis.

Put $u = Q^f$, $w = Q^s$. The quantum relations used in Lemma 2.13 give $p = S \star F + wS$, where p is the fixed point class. In the fixed cohomology basis $(1, F, S, p)$, the horizontal equation is

$$z^2 \partial_z y = (U + zD)y, \quad U = \begin{pmatrix} 0 & 0 & 2u & 3uw \\ 3 & 0 & 0 & 2u \\ 2 & w & -w & 0 \\ 0 & 2 & 1 & 0 \end{pmatrix}, \quad D = \text{diag}(1, 0, 0, -1).$$

The basis $(1, F, S, S \star F)$ used for the earlier Euler quartic is related by $P = I - wE_{34}$. Its grading is $P^{-1}DP$, with entry $(3, 4) = -w$; it must not be replaced by a diagonal grading in that basis. Direct multiplication gives

$$\det(xI - U) = x^4 + wx^3 - 8ux^2 - 36uwx + 16u^2 - 27uw^2, \\ \text{disc}_x \det(xI - U) = -u^2w^2(256u + 27w^2)^3.$$

At the stated specialization this factors as $(x + 18t)^2(x^2 - 20tx + 612t^2)$. For $L_t = \text{diag}(1, t^{-1}, t^{-1}, t^{-2})$, one has $U(-27t^2, 16t) = tL_t U(-27, 16)L_t^{-1}$ and $L_t D = DL_t$. These are changes of coefficients, independent of z , so it suffices to compute the residue at $t = 1$.

Write $U_0 = U(-27, 16)$. The projector onto the double eigenspace is

$$E = \frac{U_0^2 - 20U_0 + 612I}{1296}, \quad E^2 = E, \quad \text{rk } E = 2, \quad (U_0 + 18I)E = 0.$$

In its basis $(3, -1/2, 1, 0)^t, (72, -9, 0, 1)^t$, the compression of D is

$$R = \begin{pmatrix} -1/18 & 4/3 \\ 1/54 & 1/18 \end{pmatrix}, \quad R^2 = \frac{1}{36}I, \quad \text{tr } R = 0.$$

Splitting the distinct leading clusters by a gauge $I + O(z)$ leaves this compressed coefficient unchanged: the first commutator has zero diagonal block against a scalar leading term. Removing the scalar irregular exponential therefore leaves a regular-singular block with residue eigenvalues $\pm 1/6$. The other clusters are rank one and have zero residue by Lemma 2.22. There is no loop ramification, so these four exponents give (2.2).

The divisors $S + F, F$ pair dually with f, s . Their independent tags give

$$256Q^f + 27(Q^s)^2 = 6912t^2(e^{2\beta} - e^\alpha) \neq 0.$$

The tagged quartic is separable over the fraction field of $\mathbf{C}[[t, \alpha, \beta]]$; all four leading blocks have rank one, hence zero primitive-sixth multiplicity. Specializing $\alpha = \beta = 0$ gives the asserted jump. $\square$

This example does not meet the graded-monomial hypothesis of the direct F_1 vanishing result and does not refute restricted Hypothesis T. Its scalar double block has zero nilpotent part, so it also lies outside the nonzero-nilpotent-block criterion of [14] and the cyclic-block persistence argument here. Faithful transport to the tagged coefficient field and preservation upon setting its tags to zero are different claims.

The collision mechanism belongs to existing caustic theory. Reyes [13, Proposition 5.3] computes the nonzero exponents $\pm(m-2)/(2m)$ along an $I_2(m)$ caustic under his Frobenius-manifold and nondegeneracy hypotheses. At $m = 3$ these are $\pm 1/6$. His strong isomonodromy theorem concerns motion along the caustic, not specialization from its complement. The calculation above establishes this F_1 example directly, without claiming the general collision phenomenon or importing the geometric hypotheses of that theorem.

The remaining target is a geometrically checkable specialization criterion for the actual center maps in Hypothesis T, including control of the permitted eigenvalue collisions. Graded monomiality alone is not such a criterion for every surface. Removing restricted Hypothesis T would still leave the reconstruction-tail Hypothesis R unresolved.

Lemma 2.19 (Divisor tagging, conditional). *Assume Hypothesis T. Let $\chi : \Lambda_T \rightarrow A$ be an actual remaining surface-center map in the class specified in Hypothesis T. If the intrinsic small even quantum connection of T has no primitive-sixth framed monodromy, neither does its specialization along χ .*

Proof. If $N_1(T) = 0$, then the numerical effective monoid is $\{0\}$, so the coefficient map is already injective and there is nothing to tag. Assume otherwise. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(T)$, and introduce formal variables $\mathbf{t} = (t_1, \dots, t_\rho)$. Define

$$\chi_{\mathbf{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d)\right).$$

This formula extends continuously from monomials to the completed Novikov ring. Indeed, strict Novikov admissibility gives $v(\chi(Q^d)) = \ell(d)$, with ℓ positive and proper on the effective numerical monoid. Hence only finitely many classes contribute below any valuation cutoff, so the tagged sum converges coefficientwise in the complete ring $A[[\mathbf{t}]]$ and defines a continuous unital homomorphism. We now prove that it is injective. Take $f = \sum_d c_d Q^d \neq 0$ and put

$$\mu = \min\{v(\chi(Q^d)) : c_d \neq 0\} = \min\{\ell(d) : c_d \neq 0\}, \quad S_\mu = \{d : c_d \neq 0, \ell(d) = \mu\}.$$

Properness in (2.9) makes S_μ finite and nonempty. Write in_v for the initial class in the associated graded. The degree- μ initial form is

$$\text{in}_\mu(\chi_{\mathbf{t}}(f)) = \sum_{d \in S_\mu} c_d \text{in}_v(\chi(Q^d)) \exp\left(\sum_i t_i (D_i \cdot d)\right) \in \text{gr}_v(A)[[\mathbf{t}]],$$

and every displayed summand coefficient is nonzero. Every term outside S_μ has valuation strictly greater than μ , so a nonzero initial form proves $\chi_{\mathbf{t}}(f) \neq 0$. This is a finite combination of distinct exponential characters. The finitely many exponent vectors are distinct integral vectors. Choose $\mathbf{a} = (a_i) \in \mathbf{Z}^\rho$ outside the finitely many integral hyperplanes on which two dot products agree, and substitute $t_i = a_i s$. The resulting exponents are distinct integers. Their derivatives at $s = 0$ form a Vandermonde matrix with nonzero integer determinant, which stays invertible over $\text{Frac}(\text{gr}_v A)$ because that field has characteristic zero. The characters are therefore linearly independent, and the initial form displayed above is nonzero.

The domain hypothesis on $\text{gr}_v(A)$ is used twice, and only twice. It provides the field $\text{Frac}(\text{gr}_v A)$ of the Vandermonde step just made, and it makes v multiplicative on A , since initial forms of nonzero elements then have nonzero product. The valuation of a monomial image is $\ell(d)$ by hypothesis (2.9); multiplicativity is what preserves that value after multiplication by the tag, a unit of valuation zero. Terms with $\ell(d) > \mu$ therefore contribute nothing in degree μ , and each $d \in S_\mu$ contributes its initial form with c_d intact. Were the associated graded to have zero divisors, v would be a filtration rather than a valuation, and the displayed degree- μ identity could fail.

Injectivity of (2) embeds the intrinsic numerical Novikov fraction field into the fraction field of $A[[t]]$, fixing $\mathbf{C}$. Therefore the module obtained by the tagged map is a scalar extension of the intrinsic small module, and their framed characteristic polynomials have the same primitive-sixth multiplicities after passage to a common algebraic closure.

On the other hand, setting $t = 0$ in the tagged family gives the χ -specialized small connection. Hypothesis T is precisely the remaining assertion that this specialization does not change the two primitive-sixth multiplicities. Combining these two statements proves the lemma. Notice that no pro-Laurent or Hahn-field framed operator is invoked: the hypothesis is attached directly to the specialization step for which it is needed. $\square$

Proposition 2.20 (Low-dimensional vanishing, conditional). *Assume Hypothesis R, and Hypothesis T only for surface centers that are neither minimal nor geometrically ruled. Let T be a point, a smooth projective curve, or a smooth projective surface, and let χ be a strictly Novikov-admissible specialization, graded-monomial when $T \cong F_1$. If T is a surface that is neither minimal nor geometrically ruled, require in addition that χ be an actual center map in the class specified in Hypothesis T. Then*

$$\nu_6(T; \chi) = 0. \quad (2.3)$$

Proof. The direct specialized cases are already covered by Proposition 2.10: this includes points and curves of positive genus through the nef-canonical case, the two projective spaces $\mathbf{P}^1, \mathbf{P}^2$, every minimal surface with nef canonical bundle, and, assuming Hypothesis R, ruled surfaces over positive-genus curves.

The rational geometrically ruled surfaces, that is the Hirzebruch surfaces F_a , are covered by Proposition 2.16, again without Hypothesis T; the graded-monomial condition on χ is used only there, and only for F_1 . Together these cases cover every minimal surface: a minimal smooth projective surface has nef canonical class, or is $\mathbf{P}^2$, or is geometrically ruled over a curve, by the Enriques–Kodaira classification [2, Chapter VI]. They also cover F_1 , which is geometrically ruled but not minimal.

It remains to treat the surfaces that are neither minimal nor geometrically ruled. Work intrinsically first, before the external specialization. Every minimal surface has vanishing intrinsic invariant: apply the direct cases just listed to the identity map of Λ_T , which is strictly Novikov-admissible with $\ell = H \cdot (-)$ for an ample H , and graded-monomial because its associated graded is the monoid algebra of the effective monoid, whose monomials are a basis. A surface that is not minimal is obtained from a minimal one by point blowups, and under Hypothesis R the blowup formula (2.8) adds only point terms, which are zero; so its intrinsic small invariant vanishes as well. For this last case we use Hypothesis T through Lemma 2.19, which carries the intrinsic vanishing to χ , proving (2.3). $\square$

Theorem 2.21 (Conditional birational invariance through dimension four). *Assume Hypothesis R, and Hypothesis T only for surface centers that are neither minimal nor geometrically ruled. If Y and Y' are birational smooth projective complex varieties of the same dimension at most four, then*

$$\nu_6(Y) = \nu_6(Y').$$

Consequently, for smooth projective threefolds,

$$Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1 \implies \nu_6(Y) = \nu_6(Y').$$

Proof. The fold $w_6 : \mathcal{L}_{\mathcal{T}_{\text{fr}}} \rightarrow \mathbf{N}$ in (2) uses the occurrence-indexed ledger construction of [14]. Proposition 2.8 supplies its projective-bundle and blowup laws, including each actual center specialization. Those specializations are strictly Novikov-admissible by the calculation following Definition 2.9 and are graded-monomial by Lemma 2.12. In dimension at most four their centers have dimension at most two, so Proposition 2.20 kills every correction. The marker-ledger theorem of [14] now gives the first assertion.

If Y, Y' are smooth threefolds and $Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1$, apply the first assertion in dimension four and (2.8):

$$2\nu_6(Y) = \nu_6(Y \times \mathbf{P}^1) = \nu_6(Y' \times \mathbf{P}^1) = 2\nu_6(Y').$$

Cancelling two in $\mathbf{N}$ proves the stabilization assertion. $\square$

Lemma 2.22 (Multiplicity-one Euler blocks). *Let T be smooth projective and consider its even quantum connection at an even bulk point over an algebraically closed coefficient field. If a generalized eigenspace of Euler multiplication has rank one, that block contributes only the eigenvalue 1 to framed regular monodromy. In particular, at the small point it contributes nothing to $\nu_6(T)$.*

Proof. Write the horizontal system at the chosen even bulk point in the standard form

$$z^2 \partial_z Y = (U - z\mu)Y,$$

where U is Euler quantum multiplication and μ is the grading operator. The Frobenius property makes U self-adjoint for the Poincaré pairing, while μ is anti-self-adjoint. Generalized eigenspaces belonging to distinct eigenvalues of the self-adjoint operator U are orthogonal, so the pairing restricts nondegenerately to the one-dimensional block. If P is its spectral projector, anti-self-adjointness on this nondegenerate line therefore gives

$$P\mu P = 0.$$

Formally split this line from the other Euler eigenspaces by the normalized gauge $g = I + zg_1 + z^2g_2 + \cdots$ whose positive coefficients are block-off-diagonal. The coefficient of z on the resulting scalar block is

$$-P\mu P + P[U, g_1]P = 0;$$

the logarithmic-derivative term of the gauge begins in order z^2 . Consequently the scalar equation has the form

$$z^2 \partial_z y = (u + z^2 h(z))y, \quad h(z) \in K[[z]],$$

for the corresponding Euler eigenvalue u . Its normalized formal solution is $e^{-u/z}v(z)$ with $v(z) \in 1 + zK[[z]]$. No fractional power of the original loop coordinate and no logarithm occurs, so the framed regular monodromy on this block is the identity. $\square$

Proposition 2.23 (Product with a projective line). *For every smooth projective variety T ,*

$$\nu_6(T \times \mathbf{P}^1) = 2\nu_6(T).$$

This statement is unconditional.

Proof. The product formula for Gromov–Witten theory [3], followed by the numerical Novikov base change of Lemma 2.5, identifies the small numerical quantum connection of $T \times \mathbf{P}^1$ with the tensor-product quantum connection of the two factors. On even cohomology this is simply $H^{\text{ev}}(T) \otimes H^{\text{ev}}(\mathbf{P}^1)$. After adjoining $q^{1/2}$ in the projective-line Novikov coefficient field, quantum multiplication by $c_1(\mathbf{P}^1) = 2H$ has the two distinct eigenvalues

$$2q^{1/2}, \quad -2q^{1/2}.$$

By Lemma 2.22, each corresponding rank-one formal factor has trivial framed regular monodromy. After a common Levelt–Turrittin ramification [11, Chapter 3], the tensor product of factors $\mathcal{E}^\varphi \otimes R$ and $\mathcal{E}^\psi \otimes S$ is $\mathcal{E}^{\varphi+\psi} \otimes (R \otimes S)$. Hence tensoring a formal factor of the quantum differential module of T with one of these rank-one projective-space factors only adds its scalar irregular exponential factor and leaves the original- z -disc regular framed operator of the T factor unchanged. The extension adjoining $q^{1/2}$ is in the coefficient field and is fixed by the z -turn. Consequently the product module has two copies of every framed cyclotomic factor of T , counted with algebraic multiplicity. This gives the formula. $\square$

Corollary 2.24. *The small even framed invariants of the two projective endpoints satisfy*

$$\nu_6(\mathbf{P}^3) = \nu_6(\mathbf{P}^4) = 0.$$

Proof. For $n = 3, 4$, the relation $H^{*(n+1)} = q$ makes multiplication by $(n+1)H$ generically semisimple after adjoining $q^{1/(n+1)}$. Every Euler block has rank one, so Lemma 2.22 gives the claim. $\square$

Proposition 2.25 (The cubic packet). *For every smooth cubic threefold X ,*

$$\nu_6(X) = 2.$$

This proposition is unconditional and uses neither Hypothesis R nor Hypothesis T.

We recall the small-even block data needed for its proof. Over $F_X = \overline{\text{Frac}} \mathbf{C}[[q]]$, put $r = (3q)^{1/2}$. The calculation in [14, Section 2] separates the two scalar blocks with leading terms $6r$ and $-6r$ from a rank-two block on $\tilde{\mathbf{S}} = (\tilde{S}_3, \tilde{S}_4)^t$ satisfying

$$z^2 \partial_z \tilde{\mathbf{S}} = \left[\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} + z \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix} + z^2 \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix} + O(z^3) \right] \tilde{\mathbf{S}}. \quad (2.4)$$

The normalized splitting gauge and its inverse use only integral powers of the original loop coordinate:

$$A(z) = I + \sum_{n \geq 1} A_n z^n, \quad A_n \text{ is block-off-diagonal.} \quad (2.5)$$

Proof. We treat the rank-two block first and return to the scalar blocks below.

For the rank-two block, seek a formal solution in the form

$$\tilde{S}_3 = z^\rho(1 + O(z)), \quad \tilde{S}_4 = cz^{\rho+1}(1 + O(z)).$$

The coefficient of $z^{\rho+1}$ in the first coordinate of (2.4), and that of $z^{\rho+2}$ in the second, give respectively

$$\rho = 2c - \frac{19}{18}, \quad c(\rho + 1) = \frac{19}{18}c - \frac{8}{81}.$$

Eliminating c gives

$$\frac{1}{2} \left(\rho + \frac{19}{18} \right) \left(\rho - \frac{1}{18} \right) = -\frac{8}{81}, \quad \text{hence} \quad \rho^2 + \rho + \frac{5}{36} = 0.$$

The quadratic formula applied to (2) gives

$$\rho = -\frac{1}{6}, -\frac{5}{6}.$$

These are the eigenvalues of the modified residue R_X ; their squared difference is $\delta^\sharp(X) = 4/9$, in agreement with [14, Section 2]. The constant matrix C adjoins only the algebraic Novikov coefficient r , and (2.5) and its inverse use only integral powers of z . Hence they preserve each exponent class in $\mathbf{Q}/\mathbf{Z}$: multiplication sends $z^\rho F_X[[z]]$ to itself, and the same holds for the inverse. Thus the two classes in (2) remain the classes in the original- z -disc frame. By (2), the framed convention above assigns to a rational exponent ρ the eigenvalue $\text{Exp}_V(\rho) = e^{2\pi i \rho}$. Therefore these two classes give

$$\text{Exp}_V\left(-\frac{1}{6}\right) = e^{-\pi i/3}, \quad \text{Exp}_V\left(-\frac{5}{6}\right) = e^{-5\pi i/3} = e^{\pi i/3}.$$

Put $R = \mathbf{C}[r, r^{-1}]$. The scalar equations have the form $z^2 \partial_z \tilde{S}_\pm = (\pm 6r + h_\pm(z)) \tilde{S}_\pm$, with $h_\pm(z) \in z^2 R[[z]]$. Their normalized formal solutions are

$$\tilde{S}_\pm = e^{\mp 6r/z} u_\pm(z), \quad u_\pm(z) \in 1 + zR[[z]].$$

After removing the exponential, $\partial_z u_\pm = z^{-2} h_\pm(z) u_\pm$ has coefficients in $R[[z]]$, which gives the asserted units. Only integral powers of z occur, so each scalar block is unramified with framed regular-monodromy eigenvalue 1.

It remains to count the rank-two solutions. Write its ansatz as $\tilde{S}_3 = z^\rho \sum_{n \geq 0} a_n z^n$ and $\tilde{S}_4 = z^{\rho+1} \sum_{n \geq 0} b_n z^n$, with $a_0 = 1$. At coefficient n , the two equations have the form

$$L_{\rho+n} \begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} \alpha_n \\ \beta_n \end{pmatrix}, \quad L_s = \begin{pmatrix} s + 19/18 & -2 \\ 8/81 & s - 1/18 \end{pmatrix}, \quad \det L_s = \left(s + \frac{1}{6}\right) \left(s + \frac{5}{6}\right).$$

Here (α_n, β_n) depends only on coefficients of lower index; for $n = 0$ it is zero, and (2) is exactly the condition that L_ρ have a kernel. The roots in (2) are distinct modulo $\mathbf{Z}$, so $L_{\rho+n}$ is invertible for every $n \geq 1$. For either root, $\ker L_\rho$ is one-dimensional and its first coordinate is nonzero; the normalization $a_0 = 1$ fixes b_0 . The recursion therefore gives one formal solution for each root and no further solution in its exponent class. The two resulting solutions are independent, and exhaust the rank-two block.

Consequently each eigenvalue in (2) has algebraic multiplicity one, while (2) contributes neither primitive sixth root. Hence

$$\nu_6(X) = 2.$$

No odd-cohomology block enters this small-even calculation, and $N_1(X) = \mathbf{Z}\ell$ makes the numerical Novikov ring $\mathbf{C}[[Q^\ell]] = \mathbf{C}[[q]]$. The proof uses the Beauville-derived reduction of [14, Section 2] and ordinary formal recursion in the original loop coordinate. It also agrees with Cai's calculation [4, pp. 5–6]. □

Corollary 2.26 (Formal-germ persistence of the cubic packet). *After extending the cubic Novikov coefficient field so that $r = (3q)^{1/2}$ is a unit, the primitive-sixth multiplicity is equal to 2 throughout the formal even bulk germ at the small hyperplane point.*

Proof. At the closed point the Euler eigenvalues are $6r, -6r, 0$, with the zero cluster a single rank-two Jordan block and the two nonzero clusters of rank one. The differences from the zero eigenvalue are units, so Proposition 2.6 applies to the zero packet and freezes its two exponent classes $-1/6, -5/6$. The two simple packets remain rank one on the formal germ and contribute no primitive sixth root by Lemma 2.22. Proposition 2.25 sets the value at the closed point equal to 2. □

Corollary 2.27 (Framed count after one product stabilization). *For every smooth cubic threefold X ,*

$$\nu_6(X \times \mathbf{P}^1) = 4.$$

This statement is unconditional.

Proof. Combine Proposition 2.23 with Proposition 2.25. $\square$

Proof of Theorem 1.1. Corollary 2.27 gives $\nu_6(X \times \mathbf{P}^1) = 4$ unconditionally. Let Y be a rational smooth projective fourfold. Then Y is birational to $\mathbf{P}^4$, so Theorem 2.21 gives $\nu_6(Y) = \nu_6(\mathbf{P}^4) = 0$ by Corollary 2.24. Since $4 \neq 0$, the fourfold $X \times \mathbf{P}^1$ is not rational. $\square$

This is the framed specialization of the common marker theorem. Its fold is finer, but its operation formulas and birational invariance are conditional. The rank-two formal-exponent fold of [14] proves the same irrationality statement without either hypothesis.

2.1 A genus-eight Fano consequence

Kuznetsov associates to every smooth prime Fano threefold V of genus eight a smooth Pfaffian cubic threefold X , together with rank-two vector bundles U on V and E^* on X , whose projectivizations are related by a flop [9, Theorems 2.17–2.18].

Corollary 2.28 (Framed count for V_{14}). *Assume Hypothesis R, and Hypothesis T only for surface centers that are neither minimal nor geometrically ruled. Then*

$$\nu_6(V) = 2.$$

Proof. The flop gives a birational map between $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$. Both projectivizations are smooth projective fourfolds. By Theorem 2.21, Proposition 2.8, and Proposition 2.25,

$$2\nu_6(V) = \nu_6(\mathbf{P}_V(U)) = \nu_6(\mathbf{P}_X(E^*)) = 2\nu_6(X) = 4.$$

Hence $\nu_6(V) = 2$. $\square$

The scope of this refinement is exactly one product stabilization. No claim is made here about any further stabilization.

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This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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